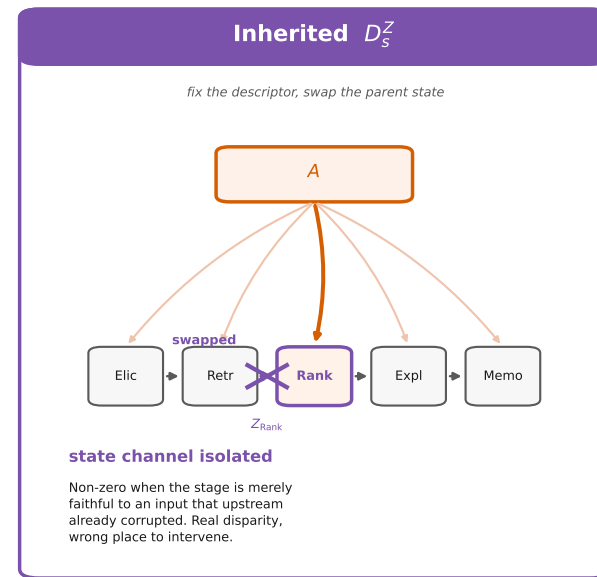
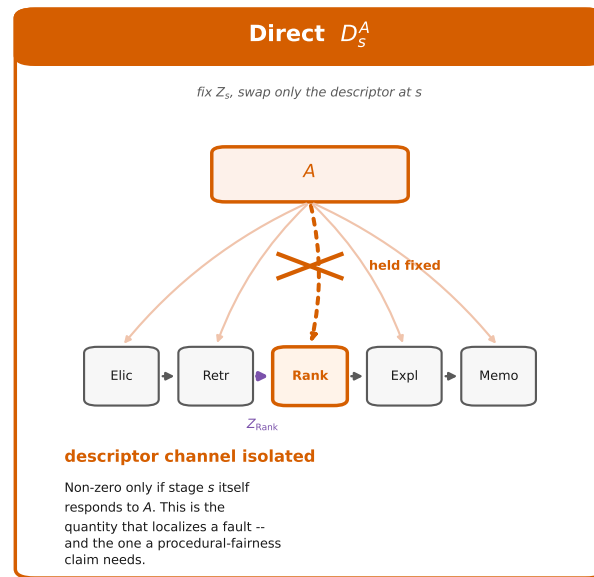
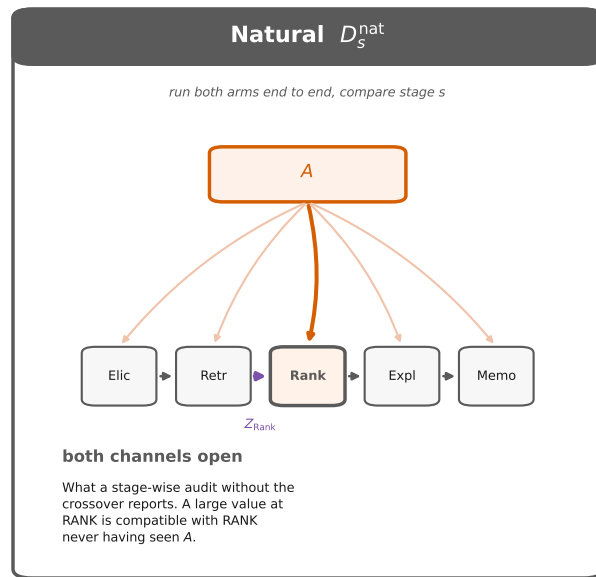


# The two SCAFF interventions as edge deletions: $D_s^A$ cuts the descriptor's direct edge into stage $s$ ; $D_s^Z$ cuts the path through its parents



## And the test that the decomposition is not just bookkeeping

Replace the implicated stage's output with the counterpart arm's and rerun every descendant. If the attribution is right, the downstream gap should collapse -- and it does: removability 0.58 at the stage  $D^A$  implicates, 0.17 at a stage it exonerates (movies; 0.59 vs 0.19 restaurants). Localization recovers the planted stage in 48/48 scenarios. In 36.5% of pairs the process is descriptor-sensitive while the final ranking stays within the endpoint-equivalence margin -- the cases an output-only audit scores as fair.